

Ex. No.: 6a)

Date: 19/02/2025

FIRST COME FIRST SERVE

Aim:

To implement First-come First-serve (FCFS) scheduling technique

Algorithm:

1. Get the number of processes from the user.
2. Read the process name and burst time.
3. Calculate the total process time.
4. Calculate the total waiting time and total turnaround time for each process.
5. Display the process name & burst time for each process.
6. Display the total waiting time, average waiting time, turnaround time

Program Code:

```
#include <stdio.h>

int main () {
    int n, i;
    float total_wt = 0, total_tat = 0;
    printf ("Enter the number of processes:");
    scanf ("%d", &n);
    int bt[n], wt[n], tat[n];
    char process[n][10];
    printf ("Enter the process names:");
    for (i=0; i<n; i++) { scanf ("%s", process[i]); }
    printf ("Enter the burst time:");
    for (i=0; i<n; i++) {
        scanf ("%d", &bt[i]);
    }
```

?

```
wt[0] = 0;
```

```
for (i=1; i<n; i++) {
```

```
    wt[i] = wt[i-1] + bt[i-1];
```

```
    total_wt += wt[i];
```

```
}
```

```
for (i=0; i<n; i++) {
```

```
    tat[i] = wt[i] + bt[i];
```

```
    total_tat += tat[i];
```

```
}
```

```
// display
```

```
printf("\n Process\t Burst time\t waiting time\t  
Turn around Time\n");
```

```
for (i=0; i<n; i++) {
```

```
    printf("%s\t%d\t%d\t%d\t\n", process[i], bt[i],
```

```
    wt[i], tat[i]);
```

```
}
```

```
printf("\n Average waiting time : %.2f\n", total_wt/n);
```

```
printf("\n Average turn around Time : %.2f\n",  
        total_tat/n);
```

```
return 0;
```

```
}
```


Sample Output:

Enter the number of processes:

3

Enter the burst time of the processes:

24 3 3

Process	Burst Time	Waiting Time	Turn Around Time
0	24	0	24
1	3	24	27
2	3	27	30

Average waiting time is: 17.0

Average Turn around Time is: 19.0

Enter the number of processes: 4

Enter process names: P1 P2 P3 P4

Enter the burst time of the processes: 5 3 8 6

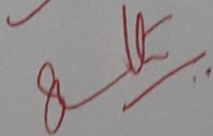
Process	Burst Time	waiting Time	Turn Around Time
			5
P1	5	0	8
P2	3	5	16
P3	8	8	22
P4	6	16	

Average waiting Time: 7.25

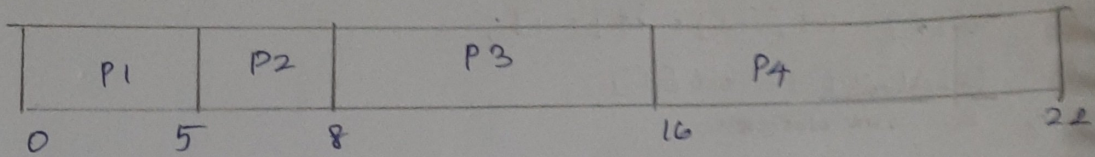
Average Turnaround Time: 12.75

Result:

Thus the code to implement First Come First serve
(FCFS) has been executed successfully



Gantt chart:



Tabulation:

Process	Burst time (ms)	Arrival time (ms)	Completion time (ms)	TAT (ms) (CT-AT)	WT (ms) (TAT-BT)
P1	5	0	5	5	0
P2	3	0	8	8	5
P3	8	0	16	16	8
P4	6	0	22	22	16

Average TAT = 12.75 ms

Average WT = 7.25 ms